# A Vision for the Siemens EDA Chair

*Bridging Industrial Reality with Academic Potentials:*

*A Life's Mission*

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# Bridging the Gap: From Theory to Industrial Scale

# Bridging the Gap: From Theory to Industrial Scale

## Academic Foundation

- **Doctoral Research Excellence**
  Gold Medal PhD: Stability analysis of delayed PDEs in large-scale circuit systems.

- **Physics-Preserving Model Reduction**
  Methodological Contributions:
  Development of structure-preserving MOR techniques enforcing conservation laws, enabling reliable **Active Reduced-Order Models** & statistically robust **Uncertainty Quantification**.

$\longrightarrow$

## Industrial Impact

- **Industrial R&D Leadership**
  Translation of theory into production-grade algorithms within **Siemens and Cadence** EDA platforms.

- **Measured Deployment Outcomes**
  Achieved **10×–100× acceleration** in post-layout verification workflows for Tier-1 semiconductor vendors.

*Research achieves impact only when it scales to industrial reality.*

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# Distinguished Research & Recognized Excellence

# Distinguished Research & Recognized Excellence

## TOP-TIER HONORS & AWARDS

**Best Transactions Paper (2013)**
*IEEE Trans. CPMT*
**Legacy:** Pioneered geometric MOR stopping criteria; now an industry standard.

**Invited State-of-the-Art (2021)**
*IEEE Journal of Microwaves*
**Impact:** Defined the global R&D roadmap for next-gen EDA tools.

**Best Paper Award (2018)**
*IEEE SPI, Brest*
Time-domain sensitivity analysis.

**Best Paper Award (2011)**
*IEEE SPI, Naples*
Projection-based MOR stopping criteria.

## PERMANENT LITERATURE

**Handbook Chapter (2020)**
*"MOR in Microelectronics"*
**In:** *Handbook on MOR*, De Gruyter.
Foundational reference for the EDA community.

## STABILITY BREAKTHROUGHS

- **2025:** Stability for active-circuit ROMs.
- **2017:** Non-linear network foundations.

**Result:** Eliminated the "unstable model" industrial bottleneck.

Research validated by over 15 years of peer-reviewed excellence.

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# Research Portfolio – Scope and Breadth

# Research Breadth: A Comprehensive EDA Portfolio

## RF, Nonlinear & Periodic Systems

- **IEEE T-MTT (2021):** PSS via DEIM-MMT **(HB)**.
- **IEEE T-MTT (2018):** NLTLs via DEIM-POD **(RF)**.
- **IEEE T-MTT (2017):** High-efficiency NLTL simulation.

## Signal & Power Integrity (SI/PI)

- **IEEE T-Adv. Packag. (2010):** System ID via $z$-Domain Basis Functions.
- **IEEE T-CPMT (2013):** Massively coupled interconnects.

## Parameterized & Statistical MOR

- **IEEE T-CPMT (2020):** MIP for DC-centric models.
- **IEEE T-EMC (2020):** PMOR for Delayed PEEC.
- **IEEE T-CPMT (2017):** Variability (NILT/PMOR).

## Sensitivity & Optimization

- **IEEE T-CPMT (2019):** Time-domain sensitivity for active networks.

”Bridging the gap between physical electromagnetics and computational efficiency by synthesizing Physics-Aware MOR and Uncertainty Quantification to deliver robust, data-driven solutions for the next generation of RF and High-Speed Systems.”

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# Strategic Research Dissemination i

## ► Major Conference and Workshop Contributions

1. **EMC Europe 2020** — Non-intrusive stochastic variability analysis with parallelism
2. **NEMO 2018** — Accurate and efficient parametric MOR for delayed *PEEC* circuits
3. **SPI 2018** — Time-domain sensitivity analysis of nonlinear circuits
4. **IMS 2018** — DEIM-based POD for nonlinear transmission-line modeling
5. **EPEPS 2017** — DEIM-based parametric MOR for DC-centric Monte Carlo analysis
6. **ICPCSI 2017** — Neural-network modeling of complex nonlinear behaviors
7. **SPI 2017** — Time-domain variability analysis of large-scale circuits with stochastic terminations
8. **EPEPS 2016** — Stability-preserving active model order reduction using generalized Lyapunov methods
9. **EPEPS 2016** — Stochastic collocation techniques for circuit variability analysis
10. **SPI 2016** — Time-domain variability analysis in high-dimensional parameter spaces

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# Strategic Research Dissemination ii

11. **IMS 2016** — Variability analysis via parameterized model order reduction
12. **EPEPS 2015** — Stability-preserving model order reduction for active networks
13. **NEMO 2015** — Model order reduction for RF application simulation
14. **EPEPS 2013** — Optimal-order estimation of reduced nonlinear macromodels
15. **SPI 2011** — Optimal-order estimation of reduced linear macromodels
16. **EPEPS 2010** — Clustering strategies for reduced-order interconnect macromodeling
17. **SPI 2008** — $z$-domain orthonormal vector fitting for high-speed module modeling

## ▶ Global Impact: Invited and Conference Talks

**Brest · Philadelphia · San Diego · Ottawa · San Jose · Naples · Austin · Avignon · Rome · Strasbourg · Ghent**

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# Selected Research Highlights: Quantitative Impact and Results

# UQ: Efficient Simulation of Nonlinear Systems

B. Nouri, E. Gad, A. Nouri, and M. Nakhla, "DC-centric parameterized reduced-order model via moment-based interpolation projection (MIP) algorithm," *IEEE Trans. CPMT*, vol. 10, no. 8, 2020.

## Challenge: The Newton-Raphson Bottleneck

- Standard Monte Carlo is too slow for SI/PI & RF.
- Iterative solvers recompute DC points for every sample.
- Evaluation of reduced nonlinear terms is the primary cost.

## Solution: Parameterized DEIM Formulation

- Developed a surrogate using **DC-centric DEIM**.
- Directly exploits structurally parameterized moments.

**MEASURED IMPACT**

**50×**

**Simulation Speed-up**

**Zero-Risk**

Tape-out Readiness

*"Moving UQ from a 'prohibitive cost' to a 'practical design-stage tool' for SI/PI & RF systems."*

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# HB: Efficient Periodic Steady-State (PSS) Analysis

B. Nouri, E. Gad, A. Nouri, and M. Nakhla, “Parameterized periodic steady-state analysis via reduced-order modelling,” *IEEE Trans. MTT*, vol. 69, no. 11, 2021.

## Problem: The ”Transformation Tax”

- Traditional solvers ”ping-pong” between Time and Frequency domains for nonlinear terms.
- FFT/IFFT operations dominate the CPU time.

## Innovation: Direct Harmonic Domain DEIM

- Evaluation of surrogates directly in the harmonic domain.
- Eliminates domain conversion overhead entirely.
- Successfully benchmarked on large-scale RF designs.

**MEASURED IMPACT**

**30×**

**Total Speedup**

**Days → Hours**

Design Cycle Reduction

*”Transforming infeasible multi-day simulations into routine, automated verification tasks for high-yield RFICs.”*

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# Stability by Construction: Active Circuit Macromodeling

G. Ghaly, E. Gad, M. Nakhla, and **B. Nouri**, "An Algorithmic Approach to Formulate Well-Conditioned Stable Reduced-Order Models of Active Circuits," *IEEE Trans. CPMT*.

## Problem: Simulation Divergence

- Active circuits yield unstable ROMs under Galerkin projection.
- Results in catastrophic failure during transient analysis.

## Innovation: Lyapunov-Based Projection

- Left-projection ($V$) satisfies **Lyapunov stability**.
- Efficient **generalized eigenproblem** avoids optimization.
- Built-in correction for numerical robustness.

**MEASURED IMPACT**

**60×**

**Total Speedup**

**Guaranteed Stability**

Numerical Robustness

*"Moving beyond 'fast' simulation to 'guaranteed' simulation for large-scale active networks."*

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# EM-Circuit Co-Simulation: Delayed PEEC Systems

B. Nouri, L. Lombardi, et al., "Parameterized model order reduction of delayed PEEC circuits," *IEEE Trans. EMC*, vol. 62, no. 3, 2020.

## Challenge: Neutral Delay DEs (NDDEs)

- High-speed retardation requires complex PEEC circuits.
- Time-stepping is bottlenecked by interpolation.
- Traditional solvers suffer from passivity crises.

## Innovation: Decoupled PMOR + NILT

- **Compact Subspaces:** Frequency-independent matching.
- **NILT Solver:** Direct Laplace-domain evaluation, eliminating time-domain interpolation overhead.

**MEASURED IMPACT**

**1000×**

**Simulation Speedup**

**NILT vs. NDDE**

Three-Order Scale Shift

*"Scaling retarded system simulation by three orders of magnitude for industrial SI/PI verification."*

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# Excellence in HQP Training & Mentorship

# Bridging the Gap Between Academia and Industry

Cultivating High-Impact Careers in Engineering Careers.

## Industrial Leadership (Siemens/Cadence)

- Onboarding and leading PhD/MSc R&D teams.
- Focus on ”Production-Ready” innovation and agile delivery.

## Academic Supervision Highlights

- **X. Deng (Carleton-2016):** Active Circuits → **Amazon AWS**
- **K. Guo (Carleton-2016):** Stochastic MOR → **AMD**
- **Y. Tao (Carleton-2016** High-dim UQ → **China**
- **Q. Sun (Carleton-2017):** DC Analysis → **Ciena**
- **A. Zhao (Carleton-2018):** Power Systems → **Microsoft**
- **G. Baggu (uOttawa-2018):** Order Selection → **Ford**
- **J. Kathrotiya (Carleton-2021):** Order Selection → **Ford**

**CAREER SUCCESS**

**Student Placements**

Bridging Industrial Reality with Academic Potential

Building the Next Generation of EDA Leaders

# Thank You

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